The final metric: X17 / IPC pairs vs neutron energy, with the γ-dark E0 channel and the trigger window

Folding formation × pair-fraction × acceptance × readout time into recorded pairs per day

Dylan Neff

15 June 2026

Internal note — informal. Drafted by Claude (Anthropic AI assistant) under D. Neff's direction; physics, numbers and text reviewed by D. Neff.

Part 0 — Summary

This note finishes the E0 thread: it turns the settled *shapes* (formation × decay, the previous note) into the **recorded** metric — X17 and IPC e^+e^- pairs per day vs neutron energy — by folding in the absolute campaign normalisation, the detector acceptance, and (the decisive new ingredient) **the trigger/readout time window**.

- **Two pair families, opposite energy ends, opposite triggers.** M1/E1 IPC pairs ride the rising radiative capture ⇒ peak at MeV ($E_n \gtrsim 100\,\mathrm{keV}$); the γ-dark E0 pairs sit **sub-keV**.
- **The flash readout cannot see E0.** At $L = 19.5\,\mathrm{m}$ the $\sim 10\,\mu\mathrm{s}$ flash readout reaches only $E_n \gtrsim 20\,\mathrm{keV}$; the E0 pairs (sub-keV, TOF $> 45\,\mu\mathrm{s}$) arrive *after* it closes. Recorded E0 in the flash window ≈ 0.
- **Only a thermal trigger records E0** — and even then it is small: with $f = \sigma_{\mathrm{E0}}/\sigma_{\mathrm{M1}} \leq 10^{-2}$, the E0 channel yields $\lesssim$3 IPC background pairs/day and $\lesssim$0.06 X17/day recorded, because the E0 capture inherits the same tiny radiative branching ($\sigma_{n\gamma}/\sigma_{np} \sim 10^{-8}$) that kills the whole sub-keV program, times f.
- **Headline (flash readout, the baseline run):** $\sim$500 IPC background pairs/day and **$\sim$6–10 X17/day** recorded (table vs anchored α_{IPC}), *all from M1/E1*; E0 adds nothing here. This reproduces the committed numbers ($\sim$190 X17 / 30-day run).
- **E0 *does* make X17** if X17 is a massive vector or scalar (it is not γ-dark to a massive boson) — but the rate is tiny because the E0 *capture* is tiny. **Bottom line: including E0 is a model-completeness improvement, not a game-changer for the measurement.**

1 The metric, assembled

The previous note settled, vs neutron energy, *which* $^4\mathrm{He}^*$ state forms (Part 1, the shape) and *whether it makes a pair* (Part 2, the height). Here we evaluate

$$N_{\mathrm{rec}}(E_n) = \underbrace{\Big[N_{\mathrm{M1/E1}}(E_n) + N_{\mathrm{E0}}(E_n; f)\Big]}_{\text{produced / pulse}} \times \underbrace{A}_{\text{acceptance}} \times \underbrace{T(E_n)}_{\text{TOF window}} ,$$

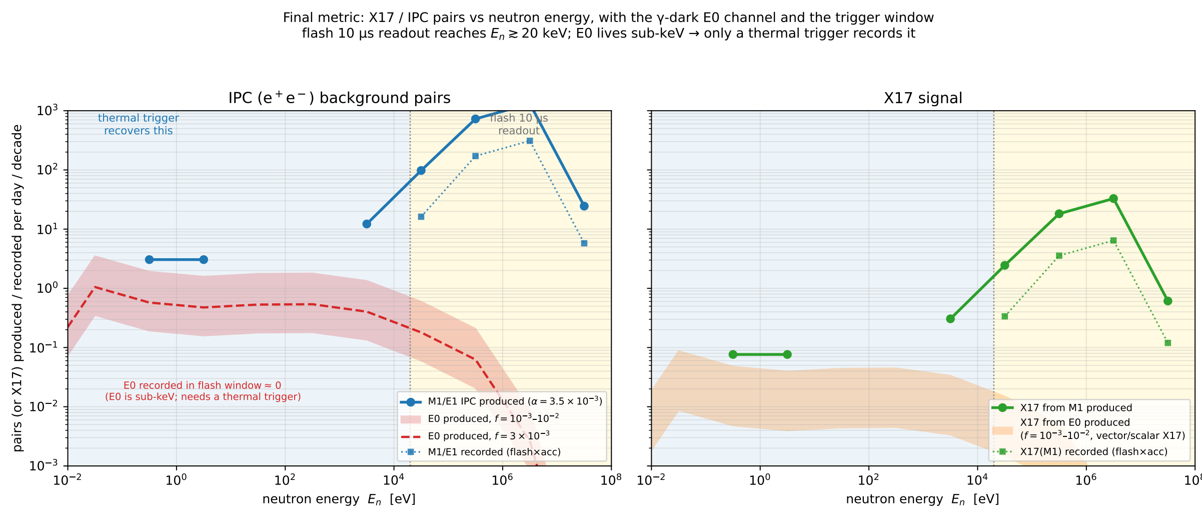
summed over decades and × pulses/day. The four ingredients, with their sources:

- **M1/E1 (term 1):** $N_{\mathrm{M1/E1}} = N_{n\gamma}^{\mathrm{direct}}(E_n)\,\alpha_{\mathrm{IPC}}$, the *directly counted* 714 $^3\mathrm{He}(n,\gamma)^4\mathrm{He}$ events per decade from the 5×10^8-neutron campaign (`mev_rates.json`), times the high-energy IPC coefficient α_{IPC}. With the table $\alpha = 2.1\times10^{-3}$ this reproduces the committed MeV-note budget exactly (32.7 X17/day produced). The $^{12}\mathrm{C}/^{8}\mathrm{Be}$-anchored $\alpha = 3.5\times10^{-3}$ is used as the default ($\times 1.67$).
- **E0 (term 2):** $N_{\mathrm{E0}} = N_{np}(E_n)\,f\,(\sigma_{n\gamma}/\sigma_{np})_{\mathrm{th}}\,g_{0^+}(E_n)$, the sub-threshold 0^+ construction of the previous note (`make_e0_pair_yield_fig.py`), carried over the bracket $f = 10^{-3}$–10^{-2}.
- **Acceptance** A**:** flat MM-double geometric acceptance from the 10^7-event at-rest `pairs_v2` study — **19.6 % (X17), 23.6 % (IPC)** (angular-resolution note). Energy-independent; the E0 monopole pair kinematics differ slightly (caveat below).
- **Trigger window** $T(E_n)$**:** $\mathrm{TOF}(E_n) = L/v$, $L = 19.5\,\mathrm{m}$. The flash readout ($\sim$10 µs) accepts $E_n \gtrsim 20\,\mathrm{keV}$; a **thermal trigger** recovers the sub-keV decades. We tabulate both.

2 The money table

per day ($\alpha_{\mathrm{IPC}} = 3.5\times10^{-3}$)	produced	recorded, flash 10 µs	recorded, flash + thermal
IPC background pairs			
M1/E1	2182	504	515
E0, $f = 10^{-3}$	1.3	0.02	0.30
E0, $f = 3\times10^{-3}$	3.9	0.05	0.91
E0, $f = 10^{-2}$	12.9	0.15	3.0
X17 signal			
from M1	54.5	10.5	10.7
from E0, $f = 10^{-3}$	0.03	0.00	0.01
from E0, $f = 10^{-2}$	0.32	0.00	0.06

With the table $\alpha = 2.1\times10^{-3}$ every M1/E1 and X17-from-M1 row scales by 0.6: produced X17 from M1 = 32.7/day, **recorded (flash) = 6.3/day** ($\approx$190 per 30-day run) — the committed baseline. The X17-from-E0 rows assume the E0 $\to$ X17 branching equals the M1 value (2.5 % per pair) as a *benchmark*; see §4.



Left: IPC background pairs/day per decade. M1/E1 (blue) peaks at MeV and is what the flash readout records (dotted, $E_n \gtrsim 20\,\text{keV}$); the E0 band (red, $f = 10^{-3}$–10^{-2}) lives sub-keV, in the region only a thermal trigger reaches. **Right:** the X17 signal — from M1 (green) and, if X17 is a massive vector/scalar, from E0 (orange band). The gold/blue shading marks the flash-window vs thermal-trigger energy regions, split at 20 keV.

3 Does E0 make X17? Yes, if X17 is massive vector or scalar

The reason $0^+ \to 0^+$ is γ-dark is specifically that **the photon is massless** — it has only transverse polarisations and cannot carry off a monopole. That is exactly why E0 instead proceeds through a *virtual* (longitudinal) photon: internal conversion and internal pair creation. **X17 is massive** (~17 MeV), so the argument that kills the photon does not kill X17. Working through the two-body $0^+ \to 0^+ + X$ selection rules:

X17 hypothesis	J^P	$0^+ \to 0^+$ emission
vector (dark photon)	1^-	**allowed** ($L = 1$)
scalar	0^+	**allowed** ($L = 0$)
pseudoscalar	0^-	forbidden
axial-vector	1^+	forbidden

So for the usually-assumed vector (dark-photon) X17 the monopole channel is open: **E0 is γ-dark but *not* X17-dark**, and could even be relatively *enhanced* (X17 is not competing against the 0.35 % IPC tail of a real γ, the same logic that makes E0 "100 % pairs"). Two honest caveats keep this from being a discovery:

1. The E0→X17 branching is a *separate* unknown — not the 2.5 % measured for the M1 line (different multipole, phase space, coupling). The table uses 2.5 % only as a placeholder.
2. Whatever the branching, the rate is bounded by the E0 *capture* rate, which is small (≤ 13 pairs/day produced even at $f = 10^{-2}$). So X17-from-E0 is $\lesssim$0.3/day produced, $\lesssim$0.06/day recorded.

4 What this means for the measurement

- **The flash-readout run is unaffected by E0.** Its ~6–10 recorded X17/day and ~500 IPC background pairs/day are entirely M1/E1; E0 contributes < 0.2/day because it is sub-keV and arrives after the window. Leaving E0 out of the generator does *not* bias the baseline high-energy analysis.
- **A thermal trigger is the only way to reach E0**, and the prize is modest: ≤ 3 extra IPC background pairs/day and ≤ 0.06 X17/day, sitting on the enormous sub-keV charged-particle load ($\sim 3 \times 10^5$ (n,p)t per pulse in the opaque gas). The statistics exist in *thermals*, but the E0 pair signal is $f \times 10^{-8}$ of them — S/B is the obstacle, not rate. A thermal trigger is still worth scoping, but E0 alone does not motivate it.
- **Generator action item is small and low-priority.** If we add E0 for completeness, it is a third pair sub-generator at the 0^+ transition energy with monopole kinematics, weighted by $N_{\text{E0}}/N_{\text{M1/E1}}$; it will not move the recorded spectra under the flash readout.

Caveats

Acceptance is the flat at-rest geometric value; the E0 monopole opening-angle distribution differs from the E1/IPC one the response was built on, so the E0 acceptance is approximate (the rate conclusion is robust to it). f remains bracketed, not pinned — a ^{4}He R-matrix normalised by the (e, e') monopole strength is still the way to a number. All MeV-region numbers are at-rest kinematics; the E_n-dependent boost ($E^* = 20.58 + \frac{3}{4}E_n$) shifts the *shapes* but not these integrated rates.

Reproducing this note

`scripts/make_e0_final_metric.py` reads `analysis/mev/mev_rates.json` and `data/He3.h5`, writes `docs/report/figs/fig_e0_final_metric.{pdf,png}` and the table numbers to `analysis/e0/final_metric.json`. Inputs trace to the MeV campaign (`mev_note.tex`), the angular-resolution acceptance (`docs/angular_resolution/`), and the E0 physics (`docs/e0_branch/`, `e0_pair_channel_note.tex`).